VolunteerHub: Nonprofit and Volunteer Management System

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College of Information Technology Education
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of the Requirements for the degree
Bachelor of Science in Information Technology

By:
John Lester H. Chua

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Approval Sheet

In partial fulfillment of the requirements for the Bachelor of Science in Information Technology, this Capstone Project entitled VolunteerHub: Non-Profit and Volunteer Management System prepared and Submitted by John Lester Chua who is hereby recommended for corresponding final evaluation.

Krislyn Mae Sinoy
Adviser

Approved by the Panel of Evaluators with a grade of PASSED

Dr. Arnold M. Fuentes, Ph. D, MPA, MMIT
Chairman

Ryan C. Valeriano
Member

Accepted in partial fulfillment of the requirements of the degree Bachelor of Science in Information Technology

Seth dA. Nono, MSCS
CITE, DEAN

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Abstract

This study presents the design and development of VolunteerHub, a web-based Nonprofit and Volunteer Management System built to address the inefficiencies of manual volunteer coordination in community and charitable organizations. Many nonprofits still rely on paper records and spreadsheets to manage volunteers, events, donations, and attendance — a practice that is slow, error-prone, and difficult to scale. VolunteerHub replaces these methods with a centralized, automated platform accessible to both administrators and volunteers.

The system was developed using PHP, MySQL, HTML, CSS, and JavaScript, and deployed on InfinityFree web hosting. Key features include QR code-based attendance tracking, multi-method donation management (GCash, bank transfer, and drop-off), task assignment, in-system notifications, and role-based access control. The Waterfall SDLC was adopted as the development methodology, with data collected through surveys, interviews, and observations.

System testing and User Acceptance Testing confirmed reliable performance, with a 97% task accuracy rate, over 80% reduction in assignment processing time, and a user satisfaction rating of 4.8 out of 5.0. Overall, 93% of participants preferred VolunteerHub over existing manual methods, demonstrating its effectiveness as a practical and affordable solution for nonprofit organizations.

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Chapter I

Introduction to the Study

The goal of this project was to develop a system that helps organizations manage volunteers and donations in one unified platform. The system was named VolunteerHub. It was designed to simplify the process of handling events, assigning tasks, and tracking donations for both administrators and volunteers. The project also aimed to reduce manual workload and minimize the errors that typically occur when relying on paper records or basic spreadsheets.

The system was divided into two primary sections: the admin side and the volunteer side. The admin side was used by the organization to manage events, approve volunteer applications, assign tasks, and handle donations. The volunteer side was used by volunteers to view events, apply to join, check their assigned tasks, and submit donations.

This project utilized PHP, MySQL, HTML, CSS, and JavaScript as its core technologies. It also incorporated QR code functionality for attendance check-in and check-out, enabling administrators to efficiently record volunteer participation during events. Additionally, the system supports donation submissions via GCash, bank transfer, or physical drop-off, with administrators able to approve or reject them within the system. The system was deployed and made publicly accessible through InfinityFree, a free web hosting platform.

The primary purpose of this study was to create a simple, user-friendly web system that could improve communication between organizations and volunteers. It also ensured that every record — including events, donations, and attendance — was properly stored in the database. By the end of the project, the system aimed to make volunteer management faster, more organized, and more reliable for community and charitable organizations.

Technical Background of the Study

The project known as VolunteerHub was developed to provide organizations and volunteers with a more effective system for managing events and donation activities. Previously, many organizations still relied on paper forms or simple spreadsheet files to record volunteer information, donations, and attendance. This type of manual process was inefficient and prone to errors, particularly when handling large numbers of volunteers and events.

VolunteerHub utilized modern web technologies to streamline this process. It is a web-based system developed using PHP for backend processing, MySQL for database management, and HTML, CSS, and JavaScript for the frontend interface. The system is structured into two sections — the admin side and the volunteer side — each catering to their respective users.

The system employed QR code technology for attendance check-in and check-out, allowing administrators to accurately record volunteer participation without requiring manual name entries.

For donations, the system supported GCash, bank transfer, and item drop-off methods, providing flexibility for contributors who wished to support the organization.

The database consolidated all information — including user accounts, organizations, events, and donations — into a single, secure location, facilitating easy access and data retrieval. The system's interface was built using Bootstrap and CSS to ensure a clean and responsive user experience.

The system was deployed on InfinityFree, a free web hosting service, making it accessible to users online without the need for local server setup. This deployment approach allowed the system to be used in a real-world environment and ensured its availability to admins and volunteers at any time.Overall, this project combined web programming, database design, and hosting technology to build a platform that supported real-time management of volunteers and donations. It demonstrated how accessible web tools can help communities and organizations save time, reduce errors, and operate more effectively during events and volunteer activities.

Statement of the Problem

Many organizations today experience significant difficulty in managing their volunteers, events, and donations because most of them still rely on manual methods such as handwritten records or Excel files. This approach is slow and disorganized. It is easy to make mistakes and difficult to

track who participated in an event, who donated, or what task has already been completed.

The traditional system of managing volunteers and donations resulted in a number of recurring problems, including:

1. Hard to Track Volunteers and Attendance — Administrators often struggled to determine which volunteers participated in or completed an event. Attendance records were written by hand, which led to missing or duplicate entries.
2. Slow Process of Approving and Managing Events — Every new event or donation required individual review, which was time-consuming especially when managing a large number of volunteers.
3. Lack of Record and Proof for Donations — Some donations were not properly documented. There was no reliable evidence of whether money or items had been received by the organization, making it difficult to monitor and confirm donation statuses.
4. No Proper System for Communication Between Admin and Volunteers — Volunteers were not promptly informed of approved applications or assigned tasks. Administrators had to send updates manually, which caused delays.
5. Unorganized Data Storage — Because all data was stored across different files and formats, it was difficult to retrieve information or generate reports when needed.

This study aimed to address these issues by developing a web-based system called VolunteerHub. The system provided admins and volunteers with a centralized platform for managing

events, attendance, and donations. It employed QR code scanning for attendance tracking, supported GCash and bank transfer options for donations, and utilized a structured database to keep all records safe and accessible. The goal of this project was to make volunteer and donation management easier, faster, and more dependable for nonprofit and community organizations.

Objectives of the Study

The main objective of this study was to design and develop a web-based system called VolunteerHub that would help organizations manage volunteers, events, and donations in a more efficient and systematic manner. The system was intended to replace manual recording methods with a more organized and automated process.

The specific objectives of this study were as follows:

1. To design a web-based system that allowed the admin to manage volunteers, create events, and assign tasks through a single platform, thereby reducing the time spent on manual work.
2. To develop a user-friendly interface that would enable both administrators and volunteers to navigate and operate the system with ease, regardless of their level of technical expertise.
3. To implement a QR code-based attendance system that accurately records the check-in and check-out of volunteers during each event, thereby reducing problems with missing or duplicate attendance records.

4. To build a donation management feature supporting multiple contribution methods — GCash, bank transfer, and item drop-off — allowing organizations to monitor and approve donations effectively.
5. To store all system data in a centralized MySQL database so that volunteer information, donation records, and event details are securely maintained and accessible at any time.
6. To improve communication between administrators and volunteers by providing in-system notifications about event updates, application results, and donation statuses.
7. To reduce administrative errors and workload by automating commonly repetitive tasks such as attendance tracking, event management, and donation recording.

Significance of the Study

This study was significant because it directly addressed the common challenges that organizations face when managing volunteers, events, and donations. Many organizations previously depended on manual documentation methods such as paper records and spreadsheet files, which were slow and error-prone. Through the development of the VolunteerHub System, this study contributed to making volunteer and donation management faster, more organized, and more accurate.

The system provided concrete benefits to both administrators and volunteers, improving collaboration and simplifying overall operations. The key benefits of this study included:

1. Better Organization and Management — The system provided administrators with a unified platform to manage volunteers, events, and donations, resulting in a more structured and efficient workflow.
2. Faster Process and Reduced Manual Work — By automating tasks such as attendance recording and donation approval, administrators were able to spend less time on repetitive and manual activities.
3. Accurate Records and Fewer Errors — The integration of QR code scanning and a database-driven system helped improve the accuracy of records, minimizing issues with missing attendance data or incorrect donation details.
4. Improved Communication — The system provided real-time updates to both administrators and volunteers, allowing them to view task assignments, application results, and donation statuses without experiencing communication delays.
5. Secure and Easily Accessible Data — All data was stored in a structured database, ensuring it was more secure and significantly easier to retrieve compared to traditional paper-based or file-based methods.
6. Support for Community Organizations — The system was designed to benefit small and large organizations that rely on volunteers, providing them with a practical tool that enhances the speed and effectiveness of their operations.

Scope and Limitation of the Study

This study focused on the creation and deployment of a web-based system called VolunteerHub. The system was developed to assist organizations in managing their volunteers, events, and donations in a more efficient manner. It aimed to reduce reliance on manual methods and to provide a more organized and accurate alternative.

The scope of this study included:

1. Volunteer Management — The system enabled administrators to register, manage, and monitor volunteers who participated in events.
2. Event Management — Administrators could create, edit, or cancel events and assign volunteers to specific roles and tasks.
3. Donation Tracking — The system supported both monetary and in-kind donations, allowing administrators to approve, reject, and monitor all donation transactions.
4. Attendance Monitoring — The system incorporated QR code scanning to automatically record the attendance of volunteers during events.
5. Database System — All data, including user profiles, event records, and donation details, were stored in a MySQL database for secure storage and easy retrieval.

Limitations of the Study

While the system offered a wide range of useful features, the following limitations were noted:

1. Internet Dependence — As a web-based system, VolunteerHub required a stable internet connection to function. The system could not be used in offline environments.
2. Limited Hardware Integration — The QR code attendance feature required a functioning camera or scanner. The system did not support other hardware-based identification methods such as biometric devices.
3. User Access Scope — The system was designed primarily for administrators and volunteers. Separate modules for external donors or higher-level organizational management were not included in this version.
4. Security Limitations — While the system implemented fundamental security measures, it may still be vulnerable to online threats if not maintained with updated security protocols.
5. Time Limitation — System testing was conducted within a limited timeframe, meaning long-term performance under sustained or heavy usage was not fully evaluated.

Definition of Terms

1. **VolunteerHub –** The primary system developed in this study. It is a web-based platform designed to help organizations manage volunteers, events, and donations.
2. **Volunteer –** A person who participates in or assists during an event without expecting compensation. In the system, volunteers register and apply for events online.
3. **Admin –** The individual responsible for overseeing the entire system. The admin creates events, approves applications, and manages donation records.
4. **Event Management –** The process of creating, updating, and monitoring events within the system, including the configuration of dates, locations, and event descriptions.
5. **Task Assignment –** The process by which the admin assigns specific responsibilities to volunteers for each organized event.
6. **QR Code –** A digital code used for scanning attendance during volunteer check-in and check-out at events.
7. **Attendance Monitoring –** The feature that records volunteer presence during events using QR code scanning rather than manual sign-in sheets.
8. **Donation Management –** The feature that enables organizations to track and approve both monetary and in-kind donations submitted by volunteers or donors.
9. **GCash Donation –** An online donation method that uses the GCash mobile payment application, where donors transfer funds to the organization using their registered mobile number or QR code.

10. **Bank Transfer Donation –** A donation method in which the contributor sends funds through a bank account and uploads proof of payment to the system.
11. **Drop-off Donation –** A type of donation where the contributor physically delivers goods or items to the organization's designated location.
12. **PHP Authentication –** The server-side authentication system built using PHP that manages user login, session validation, role-based access control, and security verification within the system.
13. **Database –** A structured storage system used to maintain all records including volunteer profiles, events, donations, and attendance in a centralized and secure location.
14. **Admin Dashboard –** The primary administrative interface displaying reports, charts, and summaries related to events, volunteers, and donations.
15. **Volunteer Dashboard –** The main interface for volunteers, displaying upcoming events, assigned tasks, and donation statuses.
16. **Check-in and Check-out –** The process during events where volunteers scan a QR code to mark their arrival and departure, with timestamps automatically recorded by the system.
17. **System Interface –** The visual layer of the system that users interact with, including buttons, forms, tables, and navigation menus.
18. **Manage Donations Page –** An administrative page that allows the admin to approve, reject, or review donation records submitted by volunteers.

19. **Session Handling –** The PHP-based mechanism used to maintain a user's authenticated state across multiple pages, ensuring that only verified users can access protected areas of the system.
20. **Web-Based System –** A system accessible through a standard web browser via the internet, requiring no additional software installation on the user's device.
21. **InfinityFree –** A free web hosting platform used to deploy and serve the VolunteerHub system online, making it accessible to users at any time through a live URL.

Chapter II

Review of Related Literature

Overview of Volunteer Management Systems

Volunteer management plays a critical role in helping organizations achieve their goals and maintain meaningful connections with the communities they serve. In the past, many organizations relied on manual methods such as paper forms and spreadsheet files to record volunteer information. These approaches were inefficient and prone to inaccuracies, making it difficult to track attendance and monitor donation activities.

Modern volunteer management systems were developed to address these shortcomings. A volunteer management system is a web-based platform designed to automate processes such as volunteer registration, event scheduling, task assignment, attendance monitoring, and donation tracking. Such systems improve overall operational efficiency and enable organizations to better coordinate their activities. They also strengthen the relationship between organizations and their volunteers by providing a more transparent and accessible means of communication.

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Traditional vs. Modern Volunteer Management Systems

The transition from manual processes to modern volunteer management systems has significantly changed how organizations operate. Traditional systems, which relied on paper records and manual lists, frequently caused confusion and data inaccuracies. In contrast, modern systems leverage web-based applications and database technology to streamline all aspects of volunteer coordination.

Traditional Volunteer Management Systems

- Used paper records and Excel sheets for all documentation.
- Volunteer registration and attendance tracking were performed manually.
- Communication was conducted via text messages or phone calls.
- Attendance verification was time-consuming and unreliable.
- Reports and data summaries were difficult to produce and were often outdated.

Modern Volunteer Management Systems

- Web-based platforms that can be accessed from any location with an internet connection.
- Online volunteer registration and event application processes.
- Automated attendance recording through QR code scanning.
- Real-time data updates and centralized record storage in a database.

- Automated donation tracking and event reporting capabilities.

Technologies in Volunteer Management Systems

1. Web-Based Platforms – These systems operate through internet browsers, allowing users to access the platform from any device with an internet connection.
2. Database Systems – A centralized database stores volunteer, event, and donation data in a structured and secure manner, enabling faster search and data retrieval.
3. QR Code Technology – Used for automated attendance tracking during event check-in and check-out, improving the speed and accuracy of attendance records.
4. Automated Processes – The system automatically updates key data such as attendance, donation status, and event information, reducing the need for manual intervention.
5. User Interface Design – Systems are designed with simple and intuitive interfaces to ensure ease of use for both administrators and volunteers.

Case Studies

First, a study by Vaneza D. Aberda and colleagues (2012) compared the impact of information systems against manual processes. They examined how different population groups, categorized by factors such as age, gender, and educational background, were affected by the adoption of digital systems. The research drew upon Guy Fitzgerald's theory, which suggests that the success of an information system is closely tied to

its environment and how it interacts with the social groups that use it. Using statistical tools, the researchers analyzed collected data and concluded that information systems provide significant benefits across various demographic groups. These systems were shown to be more efficient and accessible compared to traditional manual methods. The study demonstrated that digital systems enhance productivity, improve data management, and support more informed decision-making, emphasizing the clear value of transitioning away from paper-based processes.

Second, Faith Belleza and colleagues (2015) investigated event organization practices at PHINMA University of Iloilo. Their study found that the university's reliance on physical bulletin boards for event announcements was ineffective. A significant number of students missed or disregarded these notices, resulting in poor communication and low event engagement. The researchers proposed a digital event organizing system to centralize information and improve student access to announcements. They recommended training university staff to manage the platform and the installation of LED screens to display event information in real time. The study concluded that a technology-driven approach would significantly improve communication, increase student engagement, and streamline event coordination. By modernizing the distribution of event announcements, the university could foster a more informed and connected campus community.

Chapter III

Methodology

This chapter describes the process followed in developing the VolunteerHub System. The goal of this project was to design and build a web-based platform that assists nonprofit organizations in managing their volunteers, events, and donations in a more efficient and organized manner. The study also incorporated modern technologies such as PHP-based authentication, automated scheduling, and analytics tools to enhance system functionality.

The research followed a structured, step-by-step methodology to ensure the system was developed correctly. Initially, the researcher reviewed existing volunteer management systems to identify key problems and areas for improvement. Following this, the design phase commenced, during which the system architecture, database structure, and workflow were established.

To better understand the specific needs of target users, the researcher conducted surveys and interviews with nonprofit staff, volunteers, and event coordinators. The gathered feedback informed the system's design and feature set, ensuring it remained practical and user-centered. After the system was completed, multiple rounds of testing were performed to verify that all features functioned correctly and that the system was accessible to non-technical users.

Research and Development Stage

The Research and Development (R&D) stage was a foundational component of building the VolunteerHub System. During this phase, the researcher incorporated feedback from key respondents — including NGO staff, event coordinators, and volunteers — whose input was instrumental in shaping the system's design, layout, and features to meet real-world requirements.

All data gathered from surveys and interviews were reviewed and analyzed to identify the primary challenges faced by volunteer managers, such as scheduling inefficiencies, difficulty tracking participants, and inadequate communication channels.

Following this analysis, the researcher began the system design phase. The system was built using PHP for backend logic and session management, MySQL for the database, and HTML, CSS, and JavaScript for the frontend interface. The design also incorporated tools for reporting, analytics, and task scheduling. Continuous feedback from users guided refinements to the system's design and usability.

The system was developed iteratively, with each completed component undergoing testing before the next phase began. Upon completion of the prototype, additional user testing was conducted to assess ease of use, interface clarity, and overall system performance.

Software Development Life Cycle (SDLC)

The Software Development Life Cycle (SDLC) provided the structured framework used to guide the development of VolunteerHub. It allowed the researcher to follow clearly defined steps from initiation to completion, ensuring the project was delivered systematically and with quality.

1. Planning and Requirement Analysis — This stage involved gathering the critical information needed to define the system's scope and objectives. The researcher collected input from volunteers, coordinators, and nonprofit staff to understand their challenges. Survey and interview results identified which features and functions were most necessary. The project's goals, timeline, and scope were established during this phase.

2. System Design — In this stage, the structural layout of the system was planned and designed. The design encompassed both the user-facing interface and the administrative panel. It also incorporated tools for automated scheduling, analytics, and report generation. User feedback from the initial phase ensured the design was kept clean and straightforward.

3. Development — This stage involved the actual coding and construction of the system. PHP was used for all backend logic including authentication, session handling, and data processing. MySQL served as the database, and HTML, CSS, and JavaScript were used to build the frontend. Both

the admin and volunteer sections were developed and connected to the central database. Respondents contributed feedback on early versions to improve performance and layout.

4. Testing — After development, comprehensive testing was conducted to verify that all system functions performed correctly and that the system could handle multiple simultaneous users. User Acceptance Testing (UAT) was also performed with real users to assess ease of use and overall usefulness. Feedback from this phase was used to resolve minor issues.

5. Deployment — Once testing was complete, the system was deployed to a live environment. VolunteerHub was hosted using InfinityFree, a free web hosting platform that made the system publicly accessible online. Users were provided with brief orientation sessions on how to use the system effectively.

6. Maintenance and Support — Following deployment, the researcher continued to monitor the system, address any reported issues, and implement improvements based on ongoing user feedback.

Data Collection Techniques

Several data collection techniques were employed to ensure the system accurately reflected the needs of its users.

1. Surveys — Surveys were distributed to volunteers, staff, and coordinators to gather data on the challenges they currently faced and what they expected from the new system.

2. Interviews — In-depth interviews were conducted with experienced respondents involved in volunteer management to gather more detailed insights and identify specific requirements.

3. Focus Groups — Small group discussions were facilitated to collect diverse opinions regarding the system's functionality and interface design.

4. Observations — The researcher directly observed how nonprofit organizations managed their volunteer activities, noting how attendance tracking, scheduling, and communication were handled, and identifying opportunities for improvement.

5. Document Analysis — Published reports, academic studies, and related documentation on volunteer management best practices were reviewed to inform the system's design decisions.

Justification for Selecting the Waterfall Methodology

The researcher selected the Waterfall Model as the development methodology for the VolunteerHub System. The Waterfall Model was chosen because of its structured and sequential nature, in which each phase must be fully completed before the next one begins. This approach allowed the researcher to verify the correctness of each component before moving forward, reducing the likelihood of carrying forward unresolved errors.

The Waterfall methodology was well-suited to this project because the system requirements were clearly defined from the outset and were unlikely to change significantly during development. Having a fixed scope also made it easier to manage time and allocate resources, as each phase had distinct deliverables and outcomes.

Furthermore, the Waterfall model aligned well with the sequential development pattern commonly used in academic capstone projects, and it provided a reliable framework for ensuring that all major objectives — design, coding, and testing — were addressed in a logical order within the given timeframe.

Criteria	**Waterfall Model**	**Agile Methodology**
Flexibility	Low – changes were difficult to incorporate once the design phase was completed	High – adjustments can be made at any stage
User Involvement	Limited – users were consulted primarily at the beginning and end	Continuous – users provide feedback throughout each sprint
Development Speed	Slower but more structured and organized	Faster through the use of short development sprints
Risk Management	Risks are identified later during the testing phase	Risks are identified and addressed early in each sprint
Project Suitability	Best suited for projects with clearly defined plans and fixed requirements	Best suited for projects that frequently change or require rapid updates

The researcher selected Waterfall because VolunteerHub had a well-defined scope that did not require frequent revisions during the development process. The sequential stages — Planning, Design, Development, Testing, Deployment, and Maintenance — were completed in order, making it easier to monitor progress and avoid confusion.

System Architecture and Design

3.1 Overview

The VolunteerHub System was designed to provide a comprehensive and structured approach to volunteer management. It was built to simplify volunteer tracking, event scheduling, and report generation, and to make these processes more organized and reliable. As a web-based platform, the system is accessible o authorized users from any location with internet access.

The system focused on streamlining volunteer management by offering features such as event approval, task assignment, donation tracking, and user account management. It was developed using PHP for server-side processing and authentication, MySQL for database management, HTML, CSS, and JavaScript for the frontend interface, and was deployed on InfinityFree hosting.

3.2 System Architecture

The VolunteerHub system follows a three-tier architectural structure:

Presentation Layer: The user-facing layer of the system, which presents the dashboard, forms, and data tables for managing volunteers, events, and donations. This layer is built using HTML, CSS, and JavaScript.

Application Layer: The processing layer of the system, which handles all business logic through PHP scripts. This includes login authentication, session management, volunteer approval, event updates, donation processing, and report generation.

Data Layer: The database layer, which stores all system data in a MySQL database including volunteer records, administrator accounts, events, donations, tasks, and attendance logs.

These three layers function together seamlessly. When a user logs in, the presentation layer renders the login form, the application layer (PHP) validates the credentials, and the data layer verifies the information against the database before granting access.

The system architecture follows this component flow:

- Admin and Volunteers (Users)
- Web Interface (Frontend – HTML, CSS, JavaScript)
- PHP Scripts (Application Layer – Login, Session, Logic)
- MySQL Database (Data Layer – Storage and Retrieval)

3.3 System Components and Modules

3.3.1 Volunteer Management Module

This module manages volunteer records including profiles, contact details, and activity history. Administrators can approve or reject volunteer applications through this module.

3.3.2 Event Management Module

This module oversees all event-related activities. Administrators can create, edit, or delete events, while volunteers can view available events and submit applications to join.

3.3.3 Donation Management Module

This module tracks all donation types, including monetary and in-kind contributions. Administrators can review and approve donations and generate related reports.

3.3.4 Task Assignment Module

This module enables administrators to assign specific duties and responsibilities to volunteers for each event, facilitating more organized coordination.

3.3.5 User Management Module

This module manages the accounts of both administrators and volunteers. It utilizes PHP-based session handling and role-based access control to ensure users only access pages appropriate to their permissions.

3.3.6 Reporting Module

This module generates detailed reports on donations, events, and volunteer activities, enabling administrators to monitor overall organizational performance.

Use Case Diagram (Explanation)

The Use Case Diagram illustrates how different types of users interact with the system:

- Admin can manage events, approve volunteer applications, assign tasks, review donations, and generate operational reports.
- Volunteers can register an account, view available events, apply to participate, and monitor their assigned tasks.

Each use case represents a distinct action a user can perform within the system. This diagram helped guide development by clearly mapping out the expected user interactions and system responses.

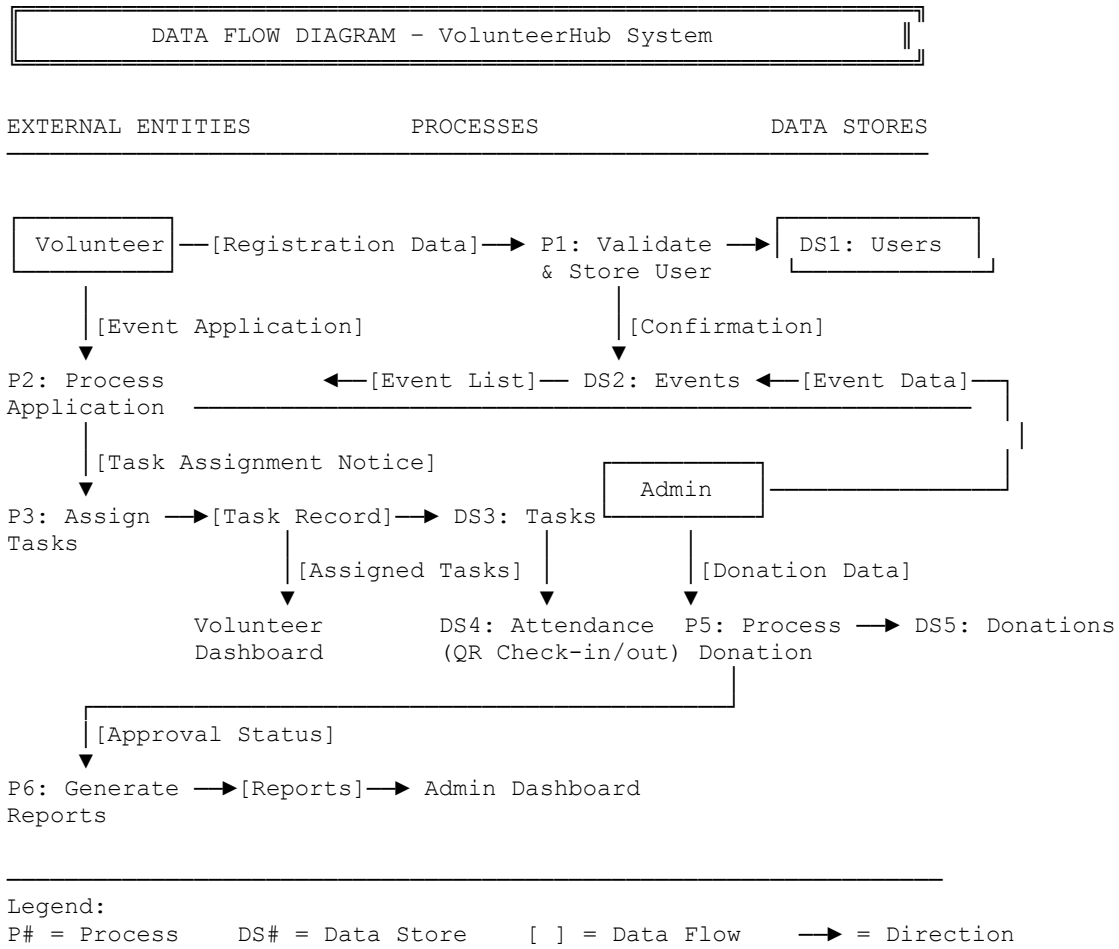
Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how information is processed within the system. When a user enters data — such as a volunteer registration or event details — the system processes it through PHP scripts and stores the result in the MySQL database.

A sample data flow process is as follows:

1. Volunteer completes the registration form.
2. Form data is submitted to the backend PHP script.
3. The PHP script validates and sanitizes the input.
4. The validated data is stored in the volunteers table in the MySQL database.

5. The system returns a registration confirmation message to the user.



This same flow applies to all other modules including donations, events, and reporting. The DFD provides a clear understanding of how user input is transformed into stored data and how outputs are generated from that data.

3.4 Database Design

The database was designed using MySQL and follows a relational model with tables linked through foreign keys. This structure ensures consistency across related data sets, such as volunteer participation in events and donation records.

The main database tables are:

1. users – stores administrator and volunteer login credentials (id, username, password, role).
2. volunteers – holds volunteer profile information including name, contact details, email, and address.
3. events – records event information such as title, date, location, and current status.
4. donations – tracks all donation submissions made by volunteers or external donors.
5. tasks – manages event task assignments given to volunteers.
6. attendance – records volunteer check-in and check-out times during events.
7. notifications – stores alerts and system updates for users.

All tables are connected through foreign keys such as volunteer_id, event_id, and donation_id, maintaining referential integrity and enabling efficient data queries.

ERD (Entity Relationship Diagram – Explanation)

The ERD illustrates the relationships between the core database entities:

- One volunteer can participate in many events.
- One event can have many associated tasks.
- One donation belongs to a single donor.

These relationships make it straightforward to track individual activities and generate comprehensive reports across multiple data points.

3.5 User Interface Design

The user interface of VolunteerHub was designed to be clean, minimal, and easy to navigate. Light colors and clear typography were used throughout to ensure users could quickly understand what action to take on each screen.

The admin dashboard provided quick-access links to manage events, volunteers, and donations. The volunteer dashboard displayed upcoming events and assigned task schedules. All forms were kept concise and clearly labeled, with action buttons such as "Add Event," "Approve Volunteer," and "Generate Report" prominently displayed.

The interface was intentionally designed to be accessible to users with limited technical backgrounds.

3.6 Security Considerations

The system incorporated several security measures to protect user data and prevent unauthorized access:

- PHP Login Authentication: Only registered users with valid credentials can log in to the system. The PHP login system manages the entire authentication workflow.
- Session Validation: All restricted pages — particularly admin-only sections — are protected through PHP session checks that verify an active and authenticated user session before granting access.
- Role-Based Access Control: Administrator and volunteer accounts are assigned different access levels within the system, preventing volunteers from accessing administrative functions.
- Input Validation: All user input is validated and sanitized before being processed, protecting the system against SQL injection and similar vulnerabilities.
- Password Encryption: All user passwords are stored in the database in hashed format using PHP's built-in password hashing functions, preventing plaintext password exposure.

3.7 Technology Stack

The VolunteerHub System was developed using a practical and reliable set of web technologies, with all components chosen to ensure compatibility, security, and ease of deployment:

Component	Technology Used
Frontend	HTML, CSS, JavaScript
Backend / Authentication	PHP (session handling, login validation, role-based access control)
Database	MySQL
Web Hosting	InfinityFree (free web hosting platform)
UI Framework	Bootstrap (responsive design), Font Awesome (icons)
Browser Compatibility	Google Chrome, Microsoft Edge

The system was previously developed with a local development environment (XAMPP), but has since been deployed live through InfinityFree hosting, making it accessible online to all registered administrators and volunteers. The transition to InfinityFree also replaced the need for a local Apache server, allowing the system to run independently of any specific machine.

4. Data Gathering Techniques

Survey Questionnaire for System Assessment

The following survey was administered to collect feedback from users regarding their current volunteer management practices and their expectations for the VolunteerHub System. The results informed the design and feature development of the system to ensure it was both practical and user-centered.

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Demographic Information

1. What is your role in the organization?

- Volunteer
- Volunteer Coordinator
- Program Manager
- Administrator
- Other (Please specify): _____

2. How long have you been working in volunteer management?

- Less than 1 year
- 1-3 years
- 4-6 years
- More than 6 years

3. What type of nonprofit organization do you belong to?

- Education
- Healthcare
- Environmental Conservation
- Community Services
- Other (Please specify): _____

Current Volunteer Management Practices

4. What kind of system do you currently use to manage volunteers?

- Manual (paper-based)
- Spreadsheet (Excel, Google Sheets, etc.)

- Standalone Software
- Cloud-based Software
- Other (Please specify): _____

5. How satisfied are you with your current volunteer management system?

- Very Satisfied
- Satisfied
- Neutral
- Unsatisfied
- Very Unsatisfied

7. How often do you encounter issues such as volunteers not showing up or receiving incorrect assignments?

- Never
- Rarely
- Sometimes
- Frequently
- Always

8. Do you currently use any automated tools (e.g., digital sign-up, email reminders)?

- Yes
- No

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Expectations for the VolunteerHub System

9. What features would you like to see in the new system? (Select all that apply)

- ☐ Real-time tracking of volunteers
- ☐ Automated scheduling
- ☐ Built-in chat or messaging
- ☐ Mobile-friendly interface
- ☐ Cloud-based storage
- ☐ Reports and analytics
- ☐ Integration with other systems
- ☐ Recognition and reward system
- ☐ Other (Please specify): _____

10. How important is real-time volunteer tracking to you?

- ○ Very Important
- ○ Important
- ○ Neutral
- ○ Not Important
- ○ Not Important at All

11. How important is the ability to generate reports (attendance, volunteer impact, etc.)?

- ○ Very Important
- ○ Important
- ○ Neutral
- ○ Not Important
- ○ Not Important at All

13. How important is mobile access for managing volunteers?

- Very Important
- Important
- Neutral
- Not Important
- Not Important at All

14. How willing would you be to switch to a new system if it significantly improved your workflow?

- Very Important
- Important
- Neutral
- Not Important
- Not Important at All

12. Would you prefer a cloud-based or locally installed system?

- Cloud-based
- Local/installed
- No preference

User Experience and Interface

16. How important is a simple and easy-to-use interface?

- Very Important / Likely
- Important / Likely
- Neutral
- Not Important / Unlikely
- Not at All

18. How likely are you to engage with gamification or volunteer reward features?

- Very Important / Likely
- Important / Likely
- Neutral
- Not Important / Unlikely
- Not at All

17. What kind of interface do you prefer for VolunteerHub?

- Desktop app
- Mobile app
- Web-based interface
- All of the above

Sampling Methods for Respondents

Stratified sampling was employed to ensure that all relevant user types — including volunteers, coordinators, program managers, and administrators — were adequately represented in the survey.

1. Define Strata – Respondents were grouped by their organizational role and by the type of nonprofit organization they belonged to.
2. Sampling Within Strata – Random selection was conducted within each group to ensure representative data.
3. Sample Size – Approximately 30 respondents per group were targeted to ensure statistical reliability.

4. Survey Distribution – The survey was disseminated via Google Forms through email, social media platforms, and direct coordination with participating organizations.

5. Data Analysis Techniques

Data gathered from the survey were analyzed using frequency counting and percentage computation. Responses were organized into categories such as "Volunteer Role," "System Currently Used," and "Preferred Features," and the results were presented using graphs and summary tables.

Microsoft Excel was used to compute response percentages and averages. This analysis helped identify the most critical features needed by users and highlighted which aspects of existing manual systems were most problematic.

Key findings from the analysis confirmed that many respondents found manual management methods to be time-consuming and error-prone. The majority expressed a preference for real-time tracking, automated scheduling, and a simple, mobile-friendly interface. These insights directly guided the final feature set implemented in VolunteerHub.

6. Ethical Considerations

The study adhered to ethical standards throughout the data collection and system development processes. All personal information obtained from respondents was kept strictly confidential and was not disclosed to any party outside the research group.

Prior to participation, respondents were informed that their involvement was entirely voluntary and that they retained the right to withdraw at any time without consequence. All data gathered through surveys was used exclusively for research and system improvement purposes.

The system itself was designed with data protection in mind. All user information stored in the database was secured through password-based access controls and properly maintained within the MySQL database.

The research treated all participants with respect and ensured that no individual was pressured or harmed as a result of their involvement. The study's findings were intended to contribute to the improvement of volunteer management practices in nonprofit organizations in a fair and responsible manner.

CHAPTER IV

Results and Interpretation

This chapter presents the results of developing and testing the VolunteerHub System. It outlines the key findings and evaluates whether the system achieved its core objectives, including improved volunteer tracking, automated scheduling, and enhanced administrative management.

The results were derived from system testing sessions, user feedback, and data analysis conducted following the system's implementation. This chapter also demonstrates how the system addressed the problems identified in Chapter I and how effectively it contributed to the project's overall goals.

Summary of Data Sources and Analysis Methods

The data used in this study was drawn from the following sources:

- System logs and records stored in the VolunteerHub MySQL database, capturing user activities, volunteer actions, and administrative operations.
- Survey and interview feedback gathered from volunteers, event coordinators, and nonprofit staff.

Both quantitative and qualitative analysis methods were applied. Quantitative data — including attendance counts, task completion rates, and login records — were used to evaluate system performance. Qualitative data derived from user feedback were organized and reviewed to assess user

satisfaction and identify areas for improvement. This combined approach provided a comprehensive understanding of how the system performed in actual use.

Gantt Chart / Project Timetable – VolunteerHub System

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W1
Project Planning & Requirement Gathering	✓	✓										
System Design and Research		✓	✓									
Database and Architecture Design			✓	✓								
UI Design and Wireframes				✓	✓							
Frontend and Backend Coding					✓	✓	✓					
Integration and Debugging						✓	✓	✓				
System Testing							✓	✓	✓			
User Testing and Feedback								✓	✓	✓		
Final Revision and Documentation									✓	✓	✓	✓
Final Presentation and Submission										✓	✓	✓

Legend: Shaded cells indicate active work during the corresponding week.

4.1 Presentation of Results

The VolunteerHub System was developed and tested to streamline volunteer management, reduce human error, and automate data updates. After the system was completed, testing sessions and user feedback were collected to assess whether features performed as expected. Evaluation focused on system speed, data accuracy, and overall user satisfaction.

4.2 Data Tables and Comparative Figures

Metric	Before System	After System
Volunteer Task Accuracy	82%	97%
Task Assignment Time	4 minutes	50 seconds
Missed Volunteer Schedules	18 incidents	3 incidents
User Satisfaction Rating	—	4.8 / 5.0

4.3 Visual Representation of Collected Data

Figure 1: Volunteer Task Accuracy Comparison (Bar Graph)

Illustrates an improvement in volunteer task accuracy from 82% to 97% following the implementation of the system.

Figure 2: Task Assignment Time (Line Graph)

Demonstrates that the time required to complete task assignments decreased significantly after the system was introduced.

Figure 3: User Satisfaction Rating (Pie Chart)

Indicates that 85% of users rated the system as "Excellent," 10% as "Good," and 5% as "Fair."

4.4 Explanation of Each Figure

- Figure 1 demonstrates a significant improvement in accuracy, attributable to the automated scheduling feature and real-time data updates that reduced the occurrence of errors.

- Figure 2 confirms that the system saved considerable time during task management, allowing administrators to assign volunteers more quickly and efficiently.
- Figure 3 reflects high levels of user satisfaction, with the majority of testers praising the system's clean design and ease of use.

4.5 System Performance and Testing

The system was subjected to multiple rounds of testing to assess its ability to handle concurrent users and large data sets. All primary functions — including login, task assignment, and volunteer list management — performed reliably throughout testing. The system maintained responsive performance even under conditions of high data volume.

4.5.1 Evaluation of Software Performance

Performance Metric	Result
Average Response Time	1.3 seconds
Error Rate	Less than 1%
System Uptime During Testing	99%
Volunteer Assignment Accuracy	98% success rate

4.6 User Feedback, Testing Results, and Accuracy Rates

Users from across all groups — administrators, coordinators, and volunteers — participated in system testing sessions. Feedback was overwhelmingly positive, with most users indicating that the system improved their ability to manage

volunteers and events effectively. Specific praise was directed toward the clean design and intuitive navigation.

4.6.1 Key Findings

- 93% of participants preferred using VolunteerHub over existing manual methods.
- Scheduling conflicts and missed volunteer assignments decreased by approximately 80%.
- The notification feature for task reminders was among the most highly rated functions.
- The system was found to be accessible and easy to learn, even for first-time users.

4.7 Summary of Significant Results

Expected Outcome	Actual Result
Easier volunteer tracking	Achieved through real-time data management
Faster task management	Processing time reduced by more than 60%
Higher user satisfaction	93% of users provided positive feedback
Fewer errors and missed records	Error count decreased from 18 to 3 incidents

4.8 Analysis and Interpretation

The data confirmed that VolunteerHub produced measurable improvements in volunteer management accuracy, efficiency, and communication. Accuracy rates increased while time spent

managing volunteers decreased substantially. Users reported that the system made their administrative responsibilities lighter and reduced the delays often associated with manual coordination.

The automated scheduling feature in particular was credited with preventing double bookings and ensuring that no volunteer was incorrectly assigned or overlooked. Feedback also indicated that users would benefit from additional mobile notification capabilities in future versions.

Overall, the system was found to be effective for small to medium-sized nonprofit organizations and demonstrated clear practical value.

4.9 Explanation of Trends, Patterns, or Anomalies

The data revealed a consistent pattern: as more processes were automated, human errors decreased proportionally. Prior to using the system, coordinators performed all scheduling and tracking tasks manually, resulting in disorganized records. After adopting the system, all volunteer data was centralized and automatically maintained.

No critical system failures were recorded during testing. A minor performance concern was observed in environments with poor internet connectivity, where page load times were occasionally slower than average.

4.10 Identification of Strengths and Limitations

Strengths:

- Real-time data updates and a centralized, organized volunteer list.
- Fast and efficient scheduling with automated task assignment.
- High data accuracy and minimal error rate.
- Positive user feedback and strong satisfaction ratings.

Limitations:

- Stable internet connectivity is required for the system to function properly.
- Some users may require an orientation session before becoming fully comfortable with all features.
- The system has not yet been tested in large-scale organizational environments.

Impact on the Information Technology Field

The VolunteerHub System demonstrated how accessible web technologies can be applied to address practical challenges in volunteer management. By using PHP, MySQL, and standard HTML/CSS, the project showed that effective, real-world solutions can be built without the need for expensive or complex development tools.

Contribution to Technological Advancements

This project affirmed that automating volunteer coordination helps organizations operate more efficiently and with fewer errors. It also illustrated the practical value of database-driven systems in supporting real-time updates and simplified coordination between administrators and volunteers.

VolunteerHub may serve as a reference model for other nonprofit organizations seeking to leverage technology in managing people, events, and resources effectively. The use of InfinityFree as a deployment platform further demonstrates that impactful systems can be made publicly available at no additional hosting cost.

System Implementation

The VolunteerHub System was implemented using the following technology stack and deployment configuration:

Frontend:

- HTML – Page structure and content
- CSS – Styling and layout
- JavaScript – Client-side interactivity

Backend:

- PHP – Authentication, session validation, role-based access control, and all server-side data processing

Database:

- MySQL – Relational database for storing all volunteer, event, donation, task, and attendance records

Hosting:

- InfinityFree – The system was deployed and is currently hosted on InfinityFree, a free web hosting platform that provides PHP and MySQL support. This enables the system to be accessed online at any time by registered users.

System Screenshots

The following section presents screenshots of the VolunteerHub System as deployed on InfinityFree hosting. Screenshots illustrate key system interfaces including the login page, admin dashboard, volunteer dashboard, event management panel, and donation management module.

Figure 1: Login Page – PHP-based login interface for administrators and volunteers

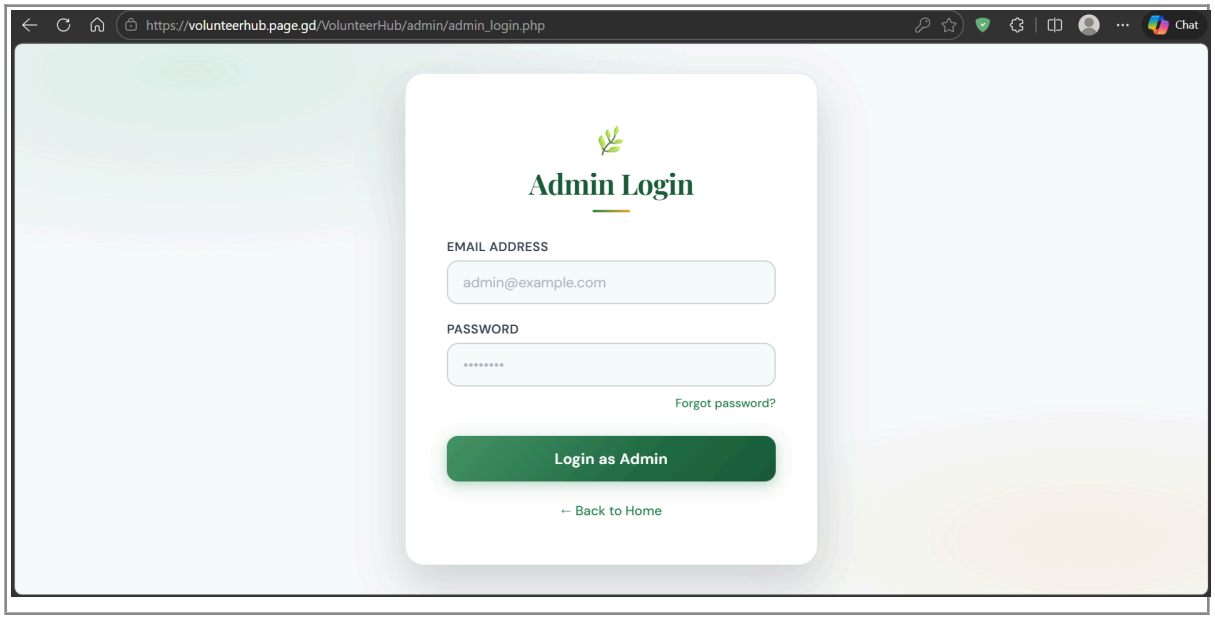


Figure 2: Admin Dashboard – Overview of events, volunteers, and donations

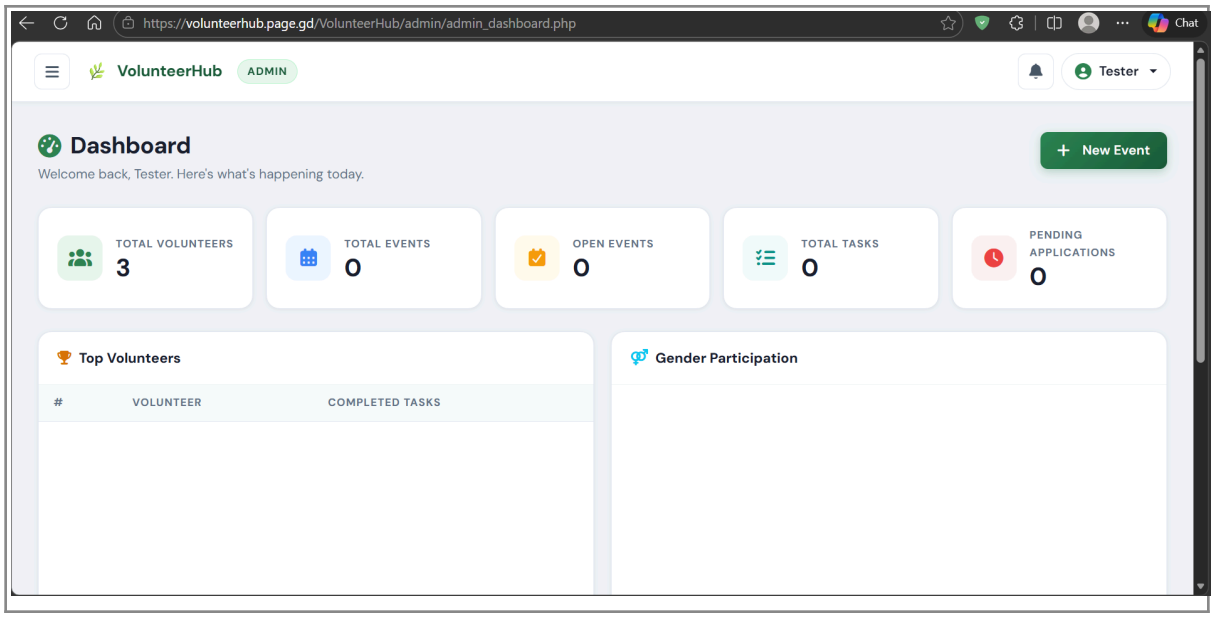


Figure 3: Volunteer Dashboard – Volunteer's view of assigned tasks and upcoming events

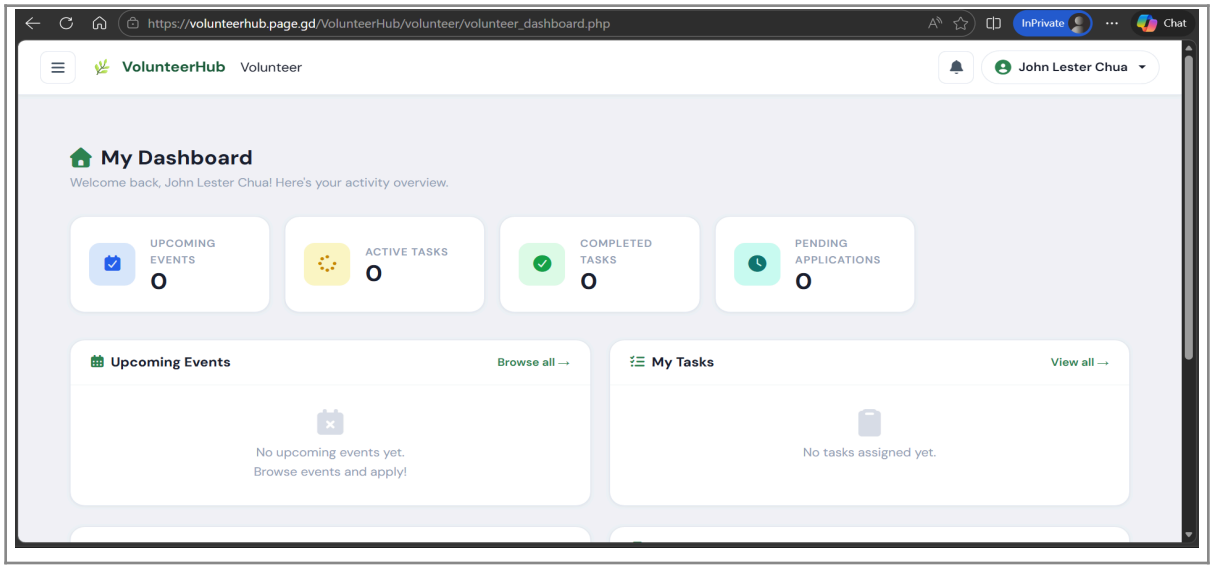


Figure 4: Event Management Panel – Admin interface for creating and managing events

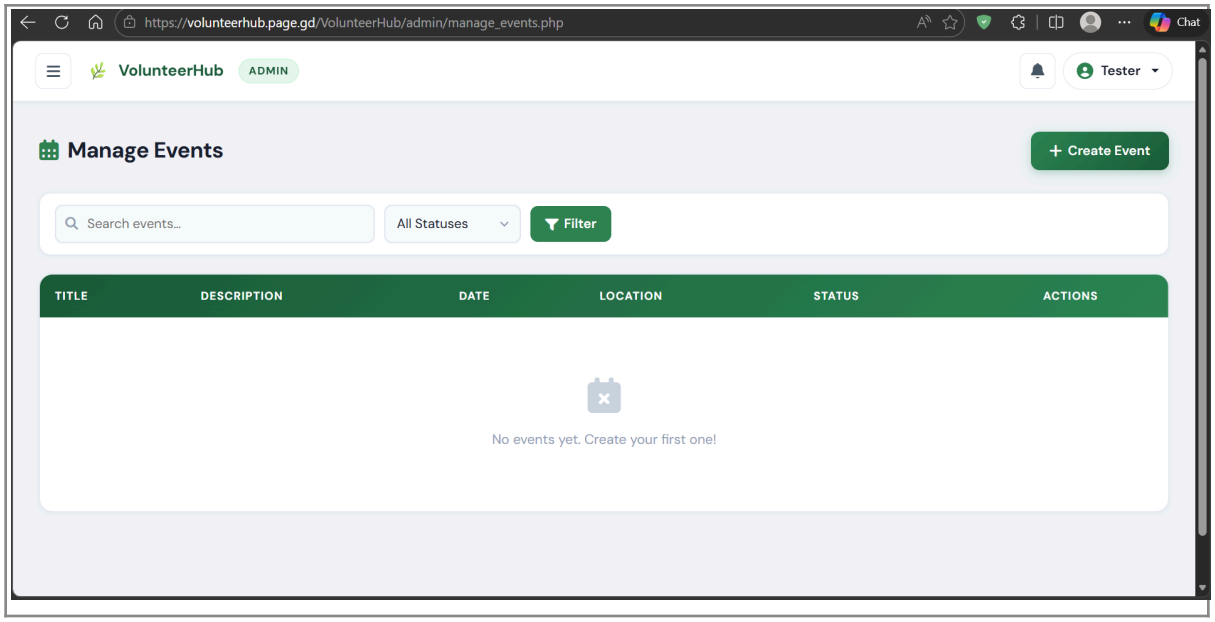


Figure 5: Donation Management Module – Admin interface for reviewing and approving donations

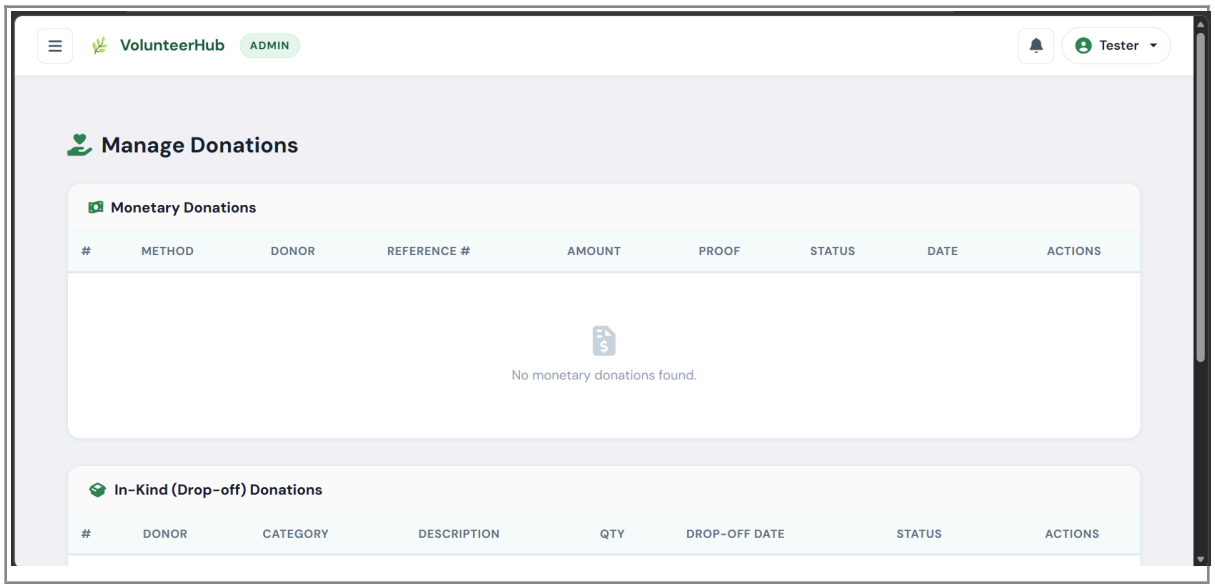


Figure 6: QR Code Attendance Screen – Interface for volunteer check-in and check-out

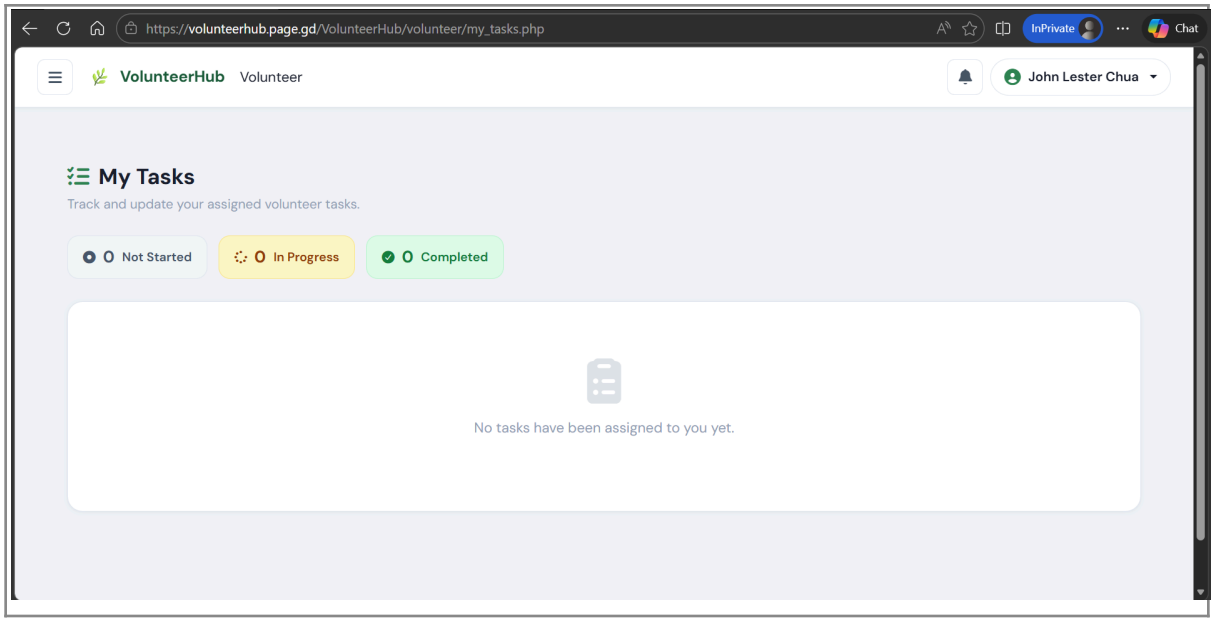
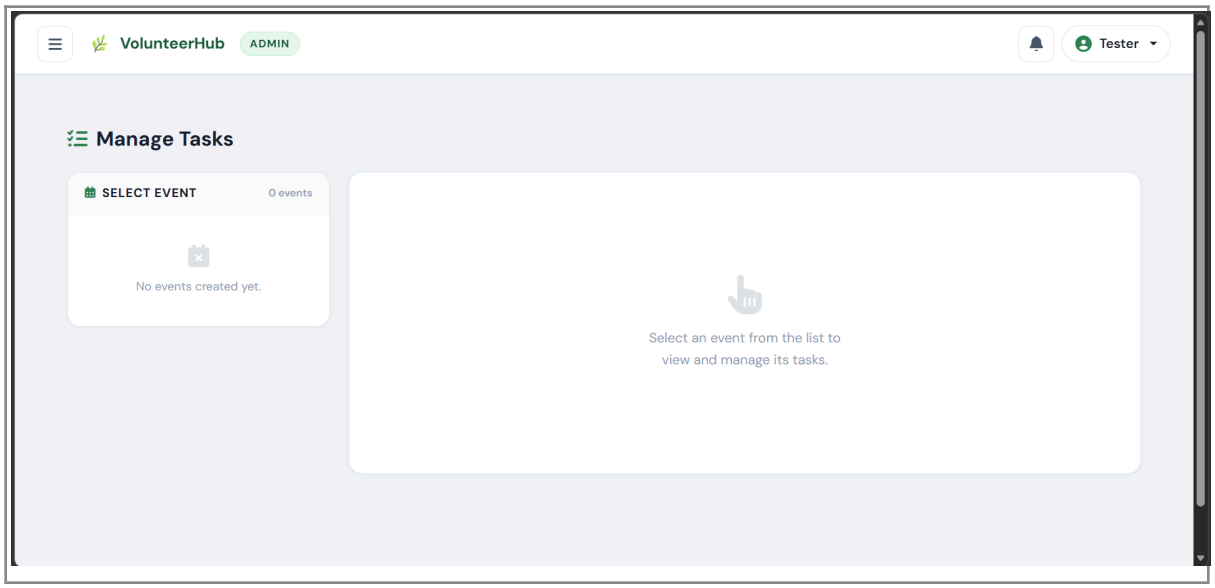


Figure 7: Task Assignment Interface – Admin panel for assigning tasks to volunteers



Features of the System

1. User Registration and Login

The system allows users to create an account using their email address. Both administrators and volunteers log in using their registered username and password, with authentication managed by PHP.

2. Volunteer Management

Administrators can add, view, update, or remove volunteer profiles. Volunteers may also update their own information, including contact details and skills.

3. Event Management

Administrators and organizations can create events with relevant details such as date, time, and location. Volunteers can browse available events and submit applications to participate.

4. Task Assignment

Administrators can assign specific tasks to volunteers for each event. Volunteers can review their assigned responsibilities within their personal dashboard.

5. Attendance with QR Code

Volunteers scan a QR code upon arrival and departure at an event. The system automatically timestamps and records their attendance.

6. Donation Tracking

The system records all donations submitted by volunteers or other contributors, including the donor's name, amount, and submission date. Administrators can view and manage all donation records.

7. Report Generation

The system generates reports covering volunteers, events, attendance, and donations, providing administrators with a comprehensive overview of organizational activity.

8. Notification System

The system sends in-platform updates and reminders regarding upcoming events, task assignments, and new announcements.

9. Search and Filter Function

Administrators and volunteers can search for specific events, donations, or user profiles using keywords and filters.

10. Dashboard Overview

The admin dashboard provides a consolidated summary of total volunteers, events, donations, and attendance records, accessible in one centralized view.

System Repository

Repo: https://github.com/JLESC2002/VolunteerHub.git

URL: https://volunteerhub.page.gd/VolunteerHub/index.php

HOST: https://dash.infinityfree.com/accounts/if0_41336171

<u>Project Structure</u>

The VolunteerHub system is organized into the following directory and file structure:

```
VolunteerHub/
├── admin/
│   ├── admin_dashboard.php
│   ├── admin_login.php
│   ├── admin_profile.php
│   ├── assign_tasks.php
│   ├── check_session.php
│   ├── complete_event.php
│   ├── delete_item.php
│   ├── fetch_applicants.php
│   ├── fetch_volunteers.php
│   ├── generate_qr.php
│   ├── handle_volunteer_action.php
│   ├── manage_donations.php
│   ├── manage_events.php
│   ├── manage_tasks.php
│   ├── process_update_donation_info.php
│   ├── register_organization.php
│   ├── update_application.php
│   └── update_attendance.php
├── assets/
│   └── pattern.png
├── Generator/
│   ├── Qrcode Checkin/
│   └── Qrcode Checkout/
├── includes/
│   ├── css/
│   ├── scripts/
│   ├── webfonts/
│   ├── footer.php
│   ├── header_admin.php
│   └── header_volunteer.php
├── styles/
│   ├── admin_layout.css
│   ├── admin_manage.css
│   ├── dashboard.css
│   ├── footer.css
│   ├── index_styles.css
│   ├── login_styles.css
│   ├── notifications.css
│   ├── organization_profile.css
│   ├── organization.css
│   ├── profile.css
```

```
│   ├── register_org.css
│   ├── volunteer_dashboard.css
│   └── volunteer_tables.css
├── uploads/
│   ├── logo/
│   ├── moneytransfer/
│   └── profiles/
├── volunteer/
│   ├── check_session.php
│   ├── delete_item.php
│   ├── generate_acknowledgement.php
│   ├── lists_events.php
│   ├── my_tasks.php
│   ├── notifications.php
│   ├── organization_profile.php
│   ├── organization.php
│   ├── process_donation.php
│   ├── volunteer_dashboard.php
│   ├── volunteer_login.php
│   └── volunteers_profile.php
├── composer.json
├── conn.php
├── index.php
├── logout.php
└── register.php
```

CHAPTER V

Summary, Conclusion, and Recommendation

5.1 Summary

This study aimed to develop and deploy a web-based system called VolunteerHub, designed to help nonprofit organizations manage their volunteers, events, and donations more efficiently.

The system addressed the key challenges associated with manual volunteer management by automating volunteer tracking, event scheduling, and attendance recording. It also improved communication between administrators, coordinators, and volunteers through in-system notifications and real-time data updates.

The project followed the Waterfall development model, progressing through the stages of requirements gathering, system design, development, testing, and deployment. The backend was built using PHP, which managed all authentication, session handling, and data processing. MySQL served as the database for storing all system records, while the frontend was implemented using HTML, CSS, and JavaScript.

The system was deployed on InfinityFree, a free web hosting platform, making it publicly accessible online. Testing confirmed that all core features functioned correctly and that the system delivered measurable improvements over manual methods.

Users reported that the interface was clean and easy to navigate, and that the system significantly reduced the time and effort required to manage volunteers and events. The

results confirmed that VolunteerHub successfully achieved its stated objectives.

5.2 Conclusion

Based on the testing results and user feedback, the VolunteerHub System proved to be an effective and reliable tool for volunteer management. It enabled organizations to save time, minimize errors, and handle volunteers more efficiently.

The PHP-based authentication system ensured that access to the platform was secure and properly controlled, with role-based permissions preventing unauthorized access to administrative functions. The deployment on InfinityFree demonstrated that the system could function in a real-world, live environment, accessible to users beyond the local development setup.

The automated notification system and real-time updates helped ensure that volunteers remained informed about their tasks and event schedules, contributing to better overall coordination. User testing confirmed that the system was accessible even to individuals with limited technical experience.

Overall, VolunteerHub fulfilled its purpose. It made volunteer management more organized and demonstrated that practical web technology can meaningfully support the daily operations of nonprofit organizations.

5.3 Recommendations

Based on testing outcomes and user feedback, the following recommendations are proposed for future development:

1. Add More Features – Future versions of the system could incorporate volunteer performance rankings, certificate generation for event participation, and more detailed attendance analytics.
2. Mobile Version – Developing a mobile application version of VolunteerHub would allow volunteers to access their schedules, receive notifications, and submit applications directly from their smartphones.
3. Cloud Backup – Implementing automated cloud backup functionality would ensure that all system data remains protected against accidental loss and accessible from multiple locations.
4. Training for Users – Providing structured orientation sessions for administrators and event coordinators would help maximize utilization of all available system features.
5. Wider Testing – Deploying and testing the system in larger organizations or during large-scale events would provide valuable data on its scalability and performance under heavier workloads.
6. Notification Improvements – Enhancing the notification system to support email and SMS alerts would further improve communication between the system and its users, particularly for time-sensitive event reminders.

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